

Lecture 10: Proof Notes for Board Work

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About These Notes: These proof notes contain detailed board work, teaching guidance, common confusions, and full proofs for the theory of strongly convex functions, optimality conditions (first-order and second-order), and line search methods. Each theorem includes color-coded steps, teaching insights, and practical implications for optimization algorithms.

Color Legend:

Green Steps	The main logical steps of a proof. Read these first to follow the flow.
Orange Keywords	Key definitions, theorems, or concepts being invoked.
Purple Results	Important conclusions, equivalences, and takeaways.
Yellow Boxes	Key algebraic manipulations or equations to write on the board.
<i>Gray Italic</i>	Teaching notes, asides, and clarifications for the instructor.

1 Theorem 1: Hessian Characterization of Strong Convexity

Theorem 1.1 (Hessian Characterization of Strong Convexity)

Let S be a nonempty open convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ be twice differentiable on S . Then f is μ -strongly convex on S if and only if the matrix

$$\nabla^2 f(\mathbf{x}) - 2\mu I_n \succeq 0 \quad \text{for all } \mathbf{x} \in S$$

Equivalently: all eigenvalues of $\nabla^2 f(\mathbf{x})$ satisfy $\lambda_i(\nabla^2 f(\mathbf{x})) \geq 2\mu$ for all $\mathbf{x} \in S$.

1.1 Proof: Forward Direction ($\Rightarrow$)

[This direction uses the definition of strong convexity. We will show that the auxiliary function $g(\mathbf{x}) = f(\mathbf{x}) - \mu\|\mathbf{x}\|^2$ is convex, and apply the second-order characterization of convexity.]

Step 1: By definition, f is μ -strongly convex if and only if the function

is convex.

$$g(\mathbf{x}) = f(\mathbf{x}) - \mu\|\mathbf{x}\|^2$$

Step 2: Compute the Hessian of g :

$$\nabla^2 g(\mathbf{x}) = \nabla^2 f(\mathbf{x}) - \nabla^2 (\mu \mathbf{x}^T \mathbf{x}) = \nabla^2 f(\mathbf{x}) - 2\mu I_n$$

[The Hessian of the quadratic form $\mu \mathbf{x}^T \mathbf{x}$ is the constant matrix $2\mu I_n$.]

Step 3: Since g is convex on the open convex set S , by the second-order characterization of convexity:

$$\nabla^2 g(\mathbf{x}) \succeq 0 \quad \text{for all } \mathbf{x} \in S$$

Step 4: Substitute the expression for $\nabla^2 g(\mathbf{x})$:

[For this direction, we assume the Hessian condition and show that $g(\mathbf{x}) = f(\mathbf{x}) - \mu\|\mathbf{x}\|^2$ is convex.]

Step 1: Assume that

$$\nabla^2 f(\mathbf{x}) - 2\mu I_n \succeq 0 \quad \text{for all } \mathbf{x} \in S$$

Step 2: Compute the Hessian of $g(\mathbf{x}) = f(\mathbf{x}) - \mu\|\mathbf{x}\|^2$:

$$\nabla^2 g(\mathbf{x}) = \nabla^2 f(\mathbf{x}) - 2\mu I_n$$

[By our assumption, this Hessian is positive semidefinite everywhere.]

Step 3: By the second-order characterization of convexity (from Lecture 6), a twice-differentiable function on an open convex set is convex if and only if its Hessian is positive semidefinite everywhere:

$$\nabla^2 g(\mathbf{x}) \succeq 0 \quad \Rightarrow \quad g \text{ is convex on } S$$

Step 4: Since $g(\mathbf{x}) = f(\mathbf{x}) - \mu\|\mathbf{x}\|^2$ is convex:

Teaching Insights:

- **Key idea:** The Hessian condition $\nabla^2 f(\mathbf{x}) - 2\mu I_n \succeq 0$ is equivalent to $\nabla^2 f(\mathbf{x}) \succeq 2\mu I_n$. This says the Hessian is “bounded below” by $2\mu I_n$.
- **Eigenvalue interpretation:** All eigenvalues of $\nabla^2 f(\mathbf{x})$ must be at least 2μ . For a quadratic, this directly determines the strong convexity parameter: $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x}$ is μ -strongly convex iff $\mu \leq \lambda_{\min}(A)/2$.
- **Connection to condition number:** The ratio $\kappa = \lambda_{\max}(A)/\lambda_{\min}(A)$ (condition number) measures how “ill-conditioned” the problem is. Strong convexity bounds help bound convergence rates of gradient methods.
- **Practical use:** This theorem provides a computational way to verify strong convexity: compute the Hessian and find its smallest eigenvalue.

1.4 Common Confusions

Common Pitfalls:

1. **“Why is it $2\mu I_n$ and not μI_n in the Hessian?”** Because the second derivative of $\mu\|\mathbf{x}\|^2 = \mu \sum_i x_i^2$ is $2\mu I_n$. The factor of 2 comes from the quadratic form.
2. **“Is strong convexity the same as strictly convex?”** No! Strict convexity is a different (and weaker in some sense) property. Strong convexity implies strict convexity, but not vice versa.
3. **“Can f be 0-strongly convex?”** No, strong convexity requires $\mu > 0$ by definition. Functions with $\mu = 0$ are merely convex, not strongly convex.

2 Theorem 2: Unique Global Optimum for Strongly Quasiconvex Functions

Theorem 2.1 (Unique Global Optimum for Strongly Quasiconvex)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be strongly quasiconvex and $S \subseteq \mathbb{R}^n$ be a nonempty convex set. If $\mathbf{x}^* \in S$ is a local minimizer of f on S , then $\mathbf{x}^*$ is the **unique global minimizer** of f on S .

Recall: A function is **strongly quasiconvex** if for all $\mathbf{x}_1, \mathbf{x}_2 \in S$ with $\mathbf{x}_1 \neq \mathbf{x}_2$ and $\lambda \in (0, 1)$:

$$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) < \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}$$

2.1 Proof

[This proof is elegant: we show that the existence of another local minimum with a smaller function value contradicts strong quasiconvexity.]

Step 1: Assume $\mathbf{x}^* \in S$ is a local minimizer of f . We want to show it is the unique global minimizer.

Step 2: Suppose for contradiction that there exists another point $\bar{\mathbf{x}} \in S$ with:

[If equality holds, we'll get a contradiction shortly. If strict inequality holds, we'll get a faster contradiction.]

Step 3: For any $\lambda \in (0, 1)$, consider the convex combination:

$$\mathbf{x}_\lambda = \lambda \mathbf{x}^* + (1 - \lambda) \bar{\mathbf{x}}$$

[$\mathbf{x}_\lambda$ lies on the segment between $\mathbf{x}^$ and $\bar{\mathbf{x}}$.]*

Step 4: Assume $f(\bar{\mathbf{x}}) < f(\mathbf{x}^*)$ (the case $f(\bar{\mathbf{x}}) = f(\mathbf{x}^*)$ is similar). Then:

$$f(\bar{\mathbf{x}}) < f(\mathbf{x}^*)$$

Step 5: By strong quasiconvexity (since $\mathbf{x}^* \neq \bar{\mathbf{x}}$ and $\lambda \in (0, 1)$):

$$f(\mathbf{x}_\lambda) < \max\{f(\mathbf{x}^*), f(\bar{\mathbf{x}})\}$$

[So the function value at $\mathbf{x}_\lambda$ is strictly less than at $\mathbf{x}^$.]*

Step 6: Choose $\lambda > 0$ small enough. Then $\mathbf{x}_\lambda$ lies in a small neighborhood of $\mathbf{x}^*$ (since $\mathbf{x}_\lambda \rightarrow \mathbf{x}^*$ as $\lambda \rightarrow 1$).

[Specifically, $\|\mathbf{x}_\lambda - \mathbf{x}^\| = (1 - \lambda)\|\mathbf{x}^* - \bar{\mathbf{x}}\| < \delta$ for sufficiently small λ , where δ is the radius of the neighborhood where $\mathbf{x}^*$ is a local min.]*

Step 7: But from Step 5, we have $f(\mathbf{x}_\lambda) < f(\mathbf{x}^*)$. This contradicts the definition of local minimizer:

$$\text{Local min at } \mathbf{x}^* \Rightarrow f(\mathbf{x}^*) \leq f(y) \text{ for all } y \in B(\mathbf{x}^*, \delta) \cap S$$

[We found a point $\mathbf{x}_\lambda$ in this neighborhood with smaller function value—contradiction!]

Teaching Insights:

- **Key difference from strictly quasiconvex:** Strictly quasiconvex guarantees uniqueness of **global** minima if one exists. Strongly quasiconvex strengthens this: a local min is automatically the unique global min.
- **Why does strong quasiconvexity matter?** It's the correct condition to upgrade from "there exists a unique global min" to "local min = global min."
- **Practical implication:** For algorithms targeting strongly quasiconvex functions, once a local minimum is found (e.g., via gradient descent), you're guaranteed to have found the global optimum!
- **Contrast with strictly quasiconvex:** Strictly quasiconvex alone only ensures uniqueness if you already know a global minimum exists—it doesn't guarantee local = global.

2.3 Common Confusions

Common Pitfalls:

1. **"Is every strongly quasiconvex function convex?"** No! Strongly quasiconvex is weaker than strictly convex. Strictly convex $\Rightarrow$ strongly quasiconvex, but not vice versa.
2. **"Why does the proof use $f(\bar{\mathbf{x}}) < f(\mathbf{x}^*)$ instead of $\leq$?"** The case $f(\bar{\mathbf{x}}) = f(\mathbf{x}^*)$ is trickier and requires a limiting argument, but the conclusion is the same: the existence of another point at the same level contradicts local minimality when strong quasiconvexity holds.
3. **"Can there be multiple local minima?"** No! If $\mathbf{x}^*$ is a local minimum and strongly quasiconvex f has another local minimum $\tilde{\mathbf{x}}$, then by this theorem, both must be global. But a global minimizer is unique, so $\mathbf{x}^* = \tilde{\mathbf{x}}$.

3 Theorem 3: First-Order Necessary Conditions (Descent Direction)

Theorem 3.1 (Descent Direction)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at $\mathbf{x}^*$, and suppose $\mathbf{d} \in \mathbb{R}^n$ satisfies

$$\nabla f(\mathbf{x}^*)^T \mathbf{d} < 0$$

Then $\mathbf{d}$ is a **descent direction** at $\mathbf{x}^*$: there exists $\delta > 0$ such that

$$f(\mathbf{x}^* + \lambda \mathbf{d}) < f(\mathbf{x}^*) \quad \text{for all } \lambda \in (0, \delta).$$

Corollary (First-Order Necessary Condition): If $\mathbf{x}^*$ is a local minimizer of f , then

$$\nabla f(\mathbf{x}^*) = \mathbf{0}$$

3.1 Proof

[This proof uses Taylor's theorem. The gradient tells us the direction of fastest increase; a negative inner product means we move in a decreasing direction.]

Step 1: Assume $\nabla f(\mathbf{x}^*)^T \mathbf{d} < 0$. By Taylor's theorem:

$$f(\mathbf{x}^* + \lambda \mathbf{d}) = f(\mathbf{x}^*) + \lambda \nabla f(\mathbf{x}^*)^T \mathbf{d} + o(\lambda)$$

where $o(\lambda)/\lambda \rightarrow 0$ as $\lambda \rightarrow 0^+$.

[Here, $o(\lambda)$ is the remainder term of order higher than λ .]

Step 2: Rearrange:

$$f(\mathbf{x}^* + \lambda \mathbf{d}) - f(\mathbf{x}^*) = \lambda \nabla f(\mathbf{x}^*)^T \mathbf{d} + o(\lambda)$$

Step 3: Let $c = \nabla f(\mathbf{x}^*)^T \mathbf{d} < 0$. Then:

$$f(\mathbf{x}^* + \lambda \mathbf{d}) - f(\mathbf{x}^*) = \lambda c + o(\lambda)$$

[The key is that c is negative and fixed; the remainder term $o(\lambda)$ becomes negligible for small λ .]

Step 4: For small enough $\lambda > 0$, the linear term dominates:

$$\lambda c + o(\lambda) < 0 \quad \Rightarrow$$

[This is possible by the definition of $o(\lambda)$.]

Step 5: More precisely, by the definition of $o(\lambda)$, there exists $\delta > 0$ such that $|o(\lambda)| < \lambda|c|/2$ for all

$$\lambda \in (0, \delta). \text{ Thus: } \lambda c + o(\lambda) < \lambda c + \frac{\lambda|c|}{2} = \lambda c + \frac{\lambda(-c)}{2} = \frac{\lambda c}{2} < 0$$

[(Recall $c < 0$, so $|c| = -c$).]

[This corollary is immediate from the contrapositive.]

Step 1: Assume $\mathbf{x}^*$ is a local minimizer. Suppose for contradiction that $\nabla f(\mathbf{x}^*) \neq \mathbf{0}$.

Step 2: Let $\mathbf{d} = -\nabla f(\mathbf{x}^*)$. Then:

$$\nabla f(\mathbf{x}^*)^T \mathbf{d} = -\nabla f(\mathbf{x}^*)^T \nabla f(\mathbf{x}^*) = -\|\nabla f(\mathbf{x}^*)\|^2 < 0$$

[Since $\nabla f(\mathbf{x}^) \neq \mathbf{0}$, the norm is positive.]*

Step 3: By Theorem 3.1, $\mathbf{d}$ is a descent direction. Thus there exists $\delta > 0$ with:

[But this contradicts the assumption that $\mathbf{x}^$ is a local minimizer!]*

Teaching Insights:

- **Intuition:** A descent direction is one where the directional derivative is negative. Taylor expansion shows that moving in such a direction locally decreases the function.
- **Geometry:** The gradient $\nabla f(\mathbf{x}^*)$ points in the direction of steepest increase. The negative gradient $-\nabla f(\mathbf{x}^*)$ is always a descent direction (if the gradient is nonzero).
- **Gradient descent:** This motivates the gradient descent algorithm: $\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \nabla f(\mathbf{x}_k)$ for $\alpha_k > 0$. The update moves in a descent direction.
- **Necessity:** The corollary says that if $\mathbf{x}^*$ is a local min, the gradient must vanish. This is a **necessary** condition—not sufficient (think saddle points).

3.4 Common Confusions

Common Pitfalls:

1. **“Does negative gradient always point to the minimum?”** Not directly—the negative gradient points downhill locally, but it might not point toward the global minimum. Multiple iterations are needed.
2. **“Can a point with $\nabla f(\mathbf{x}) = \mathbf{0}$ be a maximum or saddle point?”** Yes! The corollary only says: local min $\Rightarrow$ gradient = 0. The converse is false. Need second-order conditions to distinguish.
3. **“Why do we need λ to be small ($0 < \lambda < \delta$)?”** Because Taylor expansion is only valid locally. Far from $\mathbf{x}^*$, higher-order terms might dominate and the function might increase again.

4 Theorem 4: Second-Order Necessary Conditions

Theorem 4.1 (Second-Order Necessary Conditions)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice differentiable at $\mathbf{x}^*$. If $\mathbf{x}^*$ is a local minimizer of f , then:

1. $\nabla f(\mathbf{x}^*) = \mathbf{0}$ (first-order condition), and
2. $\nabla^2 f(\mathbf{x}^*) \succeq \mathbf{0}$ (Hessian is positive semidefinite).

4.1 Proof

[The first condition follows from Theorem 3. The second uses a second-order Taylor expansion. At a local minimum, the function cannot decrease in any direction—so the quadratic approximation must be non-negative.]

Step 1: By Theorem 3 (first-order necessary condition), since $\mathbf{x}^*$ is a local minimizer:

Step 2: By Taylor's theorem (second-order expansion):

$$f(\mathbf{x}^* + \lambda \mathbf{d}) = f(\mathbf{x}^*) + \lambda \nabla f(\mathbf{x}^*)^T \mathbf{d} +$$

where $o(\lambda^2)/\lambda^2 \rightarrow 0$ as $\lambda \rightarrow 0$.

Step 3: Substitute $\nabla f(\mathbf{x}^*) = \mathbf{0}$:

$$f(\mathbf{x}^* + \lambda \mathbf{d}) = f(\mathbf{x}^*) + \frac{\lambda^2}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} + o(\lambda^2)$$

Step 4: Since $\mathbf{x}^*$ is a local minimizer, for all $\mathbf{d} \in \mathbb{R}^n$ and sufficiently small $\lambda > 0$:

Step 5: Thus:

$$\frac{\lambda^2}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} + o(\lambda^2) \geq 0$$

Step 6: Divide by $\lambda^2/2$ and take $\lambda \rightarrow 0^+$. The remainder term $o(\lambda^2)/\lambda^2 \rightarrow 0$, leaving:

[This is the definition of positive semidefiniteness of the Hessian.]

Teaching Insights:

- **Why “necessary” not “sufficient”?** These conditions are necessary for local minima but not sufficient. A saddle point can satisfy both conditions! Example: $f(x_1, x_2) = x_1^2 - x_2^2$ at $(0, 0)$ has $\nabla f = 0$ and $\nabla^2 f$ semidefinite (eigenvalues $1, -1$ —indefinite). So not positive semidefinite.
- **Positive semidefinite vs. definite:** Semidefinite (eigenvalues ≥ 0) is necessary. Strict positive definiteness (eigenvalues > 0) is stronger and will be sufficient (see Theorem 5).
- **Geometric intuition:** At a local min, the function curves upward (or is flat) in every direction. The Hessian captures the curvature; it must be non-negative in all directions.
- **Board presentation:** The Taylor expansion is the key tool. Show clearly how the first-order term vanishes and the quadratic term dominates.

4.3 Common Confusions

Common Pitfalls:

1. **“If the Hessian is positive semidefinite, is it a minimum?”** Not necessarily! Need **strict** positive definiteness for sufficiency. Semidefinite is necessary but not sufficient.
2. **“What if the Hessian is indefinite?”** Then the point is definitely not a local minimum. Example: $f(x, y) = x^2 - y^2$ at $(0, 0)$ has indefinite Hessian and is a saddle point.
3. **“Can eigenvalues be zero?”** Yes, for positive semidefinite. Example: $f(x, y) = x^2$ has Hessian with eigenvalues 2, 0. The function doesn’t curve in the y -direction.

5 Theorem 5: Second-Order Sufficient Conditions

Theorem 5.1 (Second-Order Sufficient Conditions)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice differentiable at $\mathbf{x}^*$. If:

1. $\nabla f(\mathbf{x}^*) = \mathbf{0}$, and
2. $\nabla^2 f(\mathbf{x}^*) \succ \mathbf{0}$ (Hessian is positive definite),

then $\mathbf{x}^*$ is a **strict local minimizer** of f .

5.1 Proof

[Now we assume strict positive definiteness of the Hessian. This will allow us to show the function decreases in a neighborhood.]

Step 1: By Taylor’s theorem (second-order):

$$f(\mathbf{x}^* + \lambda \mathbf{d}) = f(\mathbf{x}^*) + \lambda \nabla f(\mathbf{x}^*)^T \mathbf{d} + \frac{\lambda^2}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} + o(\lambda^2)$$

Step 2: Substitute $\nabla f(\mathbf{x}^*) = \mathbf{0}$:

$$f(\mathbf{x}^* + \lambda \mathbf{d}) = f(\mathbf{x}^*) + \frac{\lambda^2}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} + o(\lambda^2)$$

Step 3: Since $\nabla^2 f(\mathbf{x}^*) \succ \mathbf{0}$ (positive definite), there exists $\lambda_{\min} > 0$ such that:

$$\mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} \geq \lambda_{\min} \|\mathbf{d}\|^2$$

[This is equivalent to saying all eigenvalues of the Hessian are at least $\lambda_{\min}$.]

Step 4: For any direction $\mathbf{d} \neq \mathbf{0}$, substitute into the Taylor expansion:

$$f(\mathbf{x}^* + \lambda \mathbf{d}) \geq f(\mathbf{x}^*) + \frac{\lambda^2}{2} \lambda_{\min} \|\mathbf{d}\|^2 + o(\lambda^2)$$

Step 5: For sufficiently small $\lambda > 0$, the remainder satisfies $|o(\lambda^2)| < \frac{\lambda^2}{4} \lambda_{\min} \|\mathbf{d}\|^2$. Thus:

[The quadratic term dominates, ensuring strict inequality.]

Step 6: Let $\delta > 0$ be small enough that the above holds for all $\|\mathbf{d}\| \leq \delta$. Then:

Teaching Insights:

- **Sufficient vs. necessary:** Theorem 5 gives **sufficient** conditions: if both hold, we’re guaranteed a strict local min. Contrast with Theorem 4 (necessary conditions).
- **Strict vs. non-strict:** The conclusion is a **strict** local minimum ($f(\mathbf{x}^* + h) > f(\mathbf{x}^*)$ for $h \neq 0$), stronger than just $\geq$. This comes from the positive *definiteness* (not just semidefiniteness).
- **Computational test:** To check: (1) compute $\nabla f(\mathbf{x}^*)$ and verify it’s zero, (2) compute $\nabla^2 f(\mathbf{x}^*)$ and check that all eigenvalues are positive (using Sylvester’s criterion or eigenvalue computation).
- **Practical gap:** These conditions are local. We still don’t know if this local min is *global*—need convexity or other properties for that.

5.3 Common Confusions

Common Pitfalls:

1. **“If these conditions are sufficient, are they also necessary?”** No! A strict local minimum might have $\nabla^2 f(\mathbf{x}^*) \succeq 0$ (semidefinite, not definite). Example: $f(x) = x^4$ at $x = 0$ is a strict min, but $f''(0) = 0$ (not positive definite).
2. **“Why positive definite and not semidefinite?”** Because the quadratic approximation must be positive in all directions. If the Hessian is only semidefinite, there’s a direction where the second-order term vanishes, and higher-order terms determine behavior.
3. **“Is strict local minimum the same as isolated minimum?”** Nearly—strict means $f(\mathbf{x}^* + h) > f(\mathbf{x}^*)$ for all small $h \neq 0$. Isolated adds that there are no other minima nearby. Strict is the right notion here.

6 Theorem 6: Line Search for Strictly Quasiconvex Objectives

Theorem 6.1 (Interval Reduction for Strictly Quasiconvex Line Search)

Let $\theta : \mathbb{R} \rightarrow \mathbb{R}$ be strictly quasiconvex over $[a, b]$. Let $\lambda, \mu \in [a, b]$ with $\lambda < \mu$. Then:

1. **Case 1:** If $\theta(\lambda) > \theta(\mu)$, then the minimum is not in $[a, \lambda]$:

$$\min_{\xi \in [a, b]} \theta(\xi) \geq \min_{\xi \in [\lambda, b]} \theta(\xi)$$

2. **Case 2:** If $\theta(\lambda) \leq \theta(\mu)$, then the minimum is not in $(\mu, b]$:

$$\min_{\xi \in [a, b]} \theta(\xi) \geq \min_{\xi \in [a, \mu]} \theta(\xi)$$

Implication: The interval of uncertainty can be shrunk to $[\lambda, b]$ in Case 1, or to $[a, \mu]$ in Case 2, without losing the optimal point.

6.1 Proof (Interval Reduction)

[The key is that strictly quasiconvex functions have level sets that are intervals. If we see the function increase from one point to another, we know the minimum isn't on the far side of the first point.]

Step 1: Proof of Case 1: Assume $\theta(\lambda) > \theta(\mu)$ with $\lambda < \mu$.

[We claim the minimum is not in $[a, \lambda]$.]

Step 2: Suppose for contradiction that $\theta^* = \min_{\xi \in [a, b]} \theta(\xi)$ is attained at some point $\xi^* \in [a, \lambda]$.

Then: $\theta(\xi^*) \leq \theta(\lambda)$ and $\theta(\xi^*) \leq \theta(\mu)$

Step 3: Since θ is strictly quasiconvex and $\lambda \neq \mu$ with $\xi^* < \lambda < \mu$, for any $\alpha \in (0, 1)$:

$$\theta(\alpha\xi^* + (1 - \alpha)\lambda) < \max\{\theta(\xi^*), \theta(\lambda)\} = \theta(\lambda)$$

[The strict inequality comes from strict quasiconvexity and the fact that $\xi^ \neq \lambda$.]*

Step 4: Choose α close to 0 so that $\alpha\xi^* + (1 - \alpha)\lambda$ is very close to λ but still less than λ . Then:

$$\theta(\alpha\xi^* + (1 - \alpha)\lambda) < \theta(\lambda)$$

[Similarly, by strict quasiconvexity applied to the segment from λ to μ .]

Step 5: But we can extend this logic: by strict quasiconvexity and the intermediate position of points, if $\theta(\xi^*) < \theta(\lambda)$, then θ cannot jump up to $\theta(\mu)$ at μ without violating strict quasiconvexity on the interval $[\xi^*, \mu]$.

[More precisely: for any point η strictly between ξ^ and μ , strict quasiconvexity ensures $\theta(\eta) < \max\{\theta(\xi^*), \theta(\mu)\}$. Since $\theta(\lambda) > \theta(\mu)$ and λ is between ξ^* and μ , we get a contradiction.]*

Step 6: Thus $\xi^* \notin [a, \lambda]$, so:

[Case 2 is symmetric.]

Step 1: Assume $\theta(\lambda) \leq \theta(\mu)$ with $\lambda < \mu$.

Step 2: Suppose $\xi^* = \arg \min_{\xi \in [a, b]} \theta(\xi)$ lies in $(\mu, b]$.

Step 3: Then $\theta(\xi^*) \leq \theta(\lambda) \leq \theta(\mu)$. By strict quasiconvexity on the segment $[\lambda, \mu]$:

for any $\beta \in (0, 1)$.

Step 4: Extend this argument to the segment $[\lambda, \xi^*]$. The minimum must lie in $[a, \mu]$:

Teaching Insights:

- **Practical use:** This theorem is the foundation for line search algorithms (golden section, fibonacci, etc.). By comparing two function values, we safely eliminate part of the interval.
- **Why strictly quasiconvex?** Regular quasiconvexity (allowing equality in the definition) would not guarantee strict interval reduction. Strict quasiconvexity ensures that flat regions force the minimum to be uniquely located.
- **Geometric intuition:** Strictly quasiconvex functions have the property that level sets are intervals. If we know $\theta(\lambda) > \theta(\mu)$, then the level set $\{\xi : \theta(\xi) \leq \theta(\lambda)\}$ cannot include points to the left of λ if it includes μ .
- **Algorithm connection:** This theorem is used repeatedly in derivative-free line search methods. Each iteration uses one or two function evaluations to shrink the interval.

6.4 Common Confusions

Common Pitfalls:

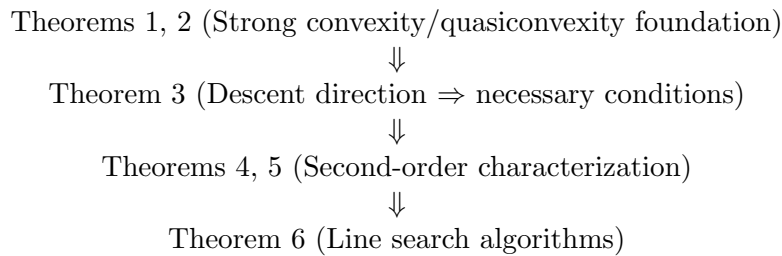
1. **“Why not use the actual minimum found so far to shrink the interval?”** The interval reduction works without computing the minimum explicitly—it just eliminates regions that cannot contain the minimum based on comparisons. This is the power of line search for expensive function evaluations.
2. **“Does this work for non-quasiconvex functions?”** No! For general functions, observing $\theta(\lambda) > \theta(\mu)$ does not guarantee anything about $[a, \lambda]$. Example: $\theta(\xi) = \sin(\xi)$ has many local minima.
3. **“What if $\theta(\lambda) = \theta(\mu)$ exactly?”** Then we’re in a boundary case. Strictly quasiconvex doesn’t forbid equal values on different points (quasiconvex does). Usually algorithms treat this as a tie-breaking situation.

7 Proof Roadmap: Summary and Connections

The following table summarizes the six theorems, their key techniques, classroom time, and logical dependencies:

Thm	Statement	Time	Priority	Key Technique
1	Hessian char. of strong convexity	6 min	High	Aux. function $g(\mathbf{x})$
2	Unique global min for strong quasi-cvx	5 min	High	Contradiction + strong quasi-cvx
3	Descent direction & FON conditions	5 min	Highest	Taylor expansion
4	Second-order necessary conditions	6 min	High	Taylor expansion (2nd order)
5	Second-order sufficient conditions	6 min	High	Positive definiteness
6	Line search for strictly quasi-cvx	8 min	Medium	Level set + contradiction

Logical Flow:



Recommended Classroom Approach:

- **Full proofs:** Theorems 1, 3, 4, 5 (core calculus-based results).
- **Proof outlines + intuition:** Theorems 2, 6 (contrapositive and geometric arguments).
- **Lecture time:** $\approx$ 36–40 minutes for all proofs with discussion.
- **Interactive moments:**
 - After Theorem 3: “Why does gradient = 0 not guarantee a minimum?”
 - After Theorem 5: “What goes wrong if the Hessian is only semidefinite?”
 - After Theorem 6: “How does this justify line search algorithms?”

8 Practice Problems and Discussion Questions

8.1 Theoretical Questions

1. (**Theorem 1 - Hessian condition**) For $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ where A is symmetric, find the relationship between the smallest eigenvalue of A and the largest μ such that f is μ -strongly convex.
2. (**Theorem 2 - Uniqueness**) If f is strongly quasiconvex on a convex set S and has a local minimum, can it have a global maximum? Discuss.
3. (**Theorem 3 - Necessary vs. sufficient**) Give an example of a function where $\nabla f(\mathbf{x}^*) = \mathbf{0}$ but $\mathbf{x}^*$ is not a local minimum (e.g., a saddle point).
4. (**Theorems 4 & 5 - Hessian test**) For the critical point of a twice-differentiable function f , explain the difference between:
 - (a) $\nabla^2 f(\mathbf{x}^*) \succeq 0$ (necessary), and
 - (b) $\nabla^2 f(\mathbf{x}^*) \succ 0$ (sufficient).

5. **(Theorem 6 - Level sets)** Explain why strictly quasiconvex functions (but not just quasiconvex) allow safe interval reduction in line search.
6. **(Strong convexity vs. strictly convex)** Is every strongly convex function strictly convex? Is every strictly convex function strongly convex? Provide proofs or counterexamples.

8.2 Computational Examples

7. **(Descent directions)** For $f(\mathbf{x}) = 2x_1^2 + x_1x_2 + 3x_2^2$, compute $\nabla f(\mathbf{x})$ at $\mathbf{x} = (1, 1)^T$ and find a descent direction.
8. **(Critical points)** Find all critical points of $f(\mathbf{x}) = x_1^3 - 3x_1x_2 + x_2^2$ and classify each using the second-order conditions.
9. **(Strong convexity)** Determine the largest $\mu > 0$ such that $f(\mathbf{x}) = 3x_1^2 + x_1x_2 + 2x_2^2$ is μ -strongly convex.
10. **(Line search)** For $\theta(\lambda) = (\lambda - 2)^2 + 1$ on $[0, 5]$, use the line search interval reduction to eliminate regions without computing the exact minimum.
11. **(Hessian test)** For $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x}$ where $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, show that the function is strongly convex and find the strong convexity constant.

8.3 Discussion Questions (For Class Engagement)

12. Theorems 3, 4, 5 form a hierarchy: necessary first-order, necessary second-order, sufficient second-order. Why is this progression important for optimization?
13. In gradient descent algorithms, we use the descent direction $\mathbf{d} = -\nabla f(\mathbf{x})$ at each step. How does Theorem 3 justify this choice?
14. Theorem 2 says strongly quasiconvex ensures uniqueness of local minima. In practice, how would you verify that a function is strongly quasiconvex?
15. Line search methods (Theorem 6) are fundamental in optimization algorithms. Describe how the interval reduction logic would be implemented in pseudocode.
16. How do Theorems 4 and 5 inform the design of optimization algorithms? What can they tell us about convergence?

9 Supplementary Remarks and Extensions

9.1 Remark: Strong Convexity and Conditioning

Strong convexity is intimately connected to problem conditioning. For a μ -strongly convex function with Lipschitz gradient (bounded Hessian), the **condition number** is $\kappa = L/\mu$, where L is the Lipschitz constant. Problems with small condition numbers (close to 1) are **well-conditioned** and converge quickly. Problems with large condition numbers are ill-conditioned. Gradient descent on a μ -strongly convex function converges exponentially fast with rate proportional to $(1 - \mu/L)^k$.

9.2 Remark: Connection to Algorithms

The optimality conditions in Theorems 3–5 are fundamental to understanding algorithm behavior:

- **Gradient descent:** Uses Theorem 3 (descent in negative gradient direction).
- **Newton’s method:** Uses the Hessian information (Theorem 5) to construct quadratic approximations.
- **Quasi-Newton methods:** Approximate the Hessian when computation is expensive.
- **Projected gradient methods:** Extend Theorem 3 to constrained optimization.

9.3 Remark: Beyond Necessary/Sufficient Conditions

In modern optimization, especially for machine learning, we often deal with non-convex objectives (e.g., neural networks). First-order and second-order conditions are still useful:

- **First-order stationarity:** $\nabla f(\mathbf{x}) = 0$ is still necessary for local minima.
- **Negative curvature escape:** If $\nabla^2 f(\mathbf{x}) \not\geq 0$ (has negative eigenvalues), we can escape saddle points.
- **Approximate stationarity:** Algorithms often find points with $\|\nabla f(\mathbf{x})\| \leq \epsilon$ for small ϵ —a relaxation of exact stationarity.

Research in non-convex optimization leverages these same theoretical tools.

9.4 Remark: Computational Complexity of Line Search

Theorem 6 shows that line search reduces the interval by a factor each iteration. For derivative-free methods:

- **Uniform search:** n evaluations reduce the interval by factor $1/(n+1)$.
- **Dichotomous/Golden section:** 2 evaluations per iteration (or 1 after the first), reduce by factor ~ 0.618 (golden section).
- **Fibonacci search:** Optimal for a fixed budget of n evaluations.

The number of iterations to achieve interval width ϵ grows logarithmically with initial width, making these methods efficient for expensive function evaluations.

These proof notes were prepared as a companion to Lecture 10 (Strongly Convex Functions, Optimality Conditions, and Line Search Methods) for ISE 5406, Optimization II, Spring 2026. For additional context, see the lecture slides and course textbooks (Bazaraa et al., Nocedal and Wright). Questions or corrections? Contact the instructor.